

## 4.3 When a quadratic form is chi-squared

A quadratic form is a weighted sum of independent noncentral  $\chi^2(1)$  pieces, and the weights are eigenvalues. It is itself chi-squared when every nonzero weight equals one. This section turns that observation into checkable conditions on  $\mathbf{A}$  and  $\Sigma$ , and shows that for the spherical case the condition is necessary as well as sufficient.

### 4.3.1 Spherical normal vectors

#### Theorem 4.3.1 — Chi-squared forms in a spherical normal vector

Let  $\mathbf{Y} \sim N_n(\boldsymbol{\mu}, \mathbf{I}_n)$ , let  $\mathbf{A}$  be symmetric, and let  $r \geq 1$  be an integer and  $\gamma \geq 0$ . Then  $\mathbf{Y}^\top \mathbf{A} \mathbf{Y} \sim \chi^2(r, \gamma)$  if and only if  $\mathbf{A}$  is idempotent with  $\text{rank}(\mathbf{A}) = r$  and  $\gamma = \boldsymbol{\mu}^\top \mathbf{A} \boldsymbol{\mu}$ .

*Proof. Sufficiency.* By Theorem 1.9.2,  $\mathbf{A} = \mathbf{Q}_1 \mathbf{Q}_1^\top$ , where the  $n \times r$  matrix  $\mathbf{Q}_1$  has orthonormal columns and  $r = \text{rank}(\mathbf{A}) = \text{tr}(\mathbf{A})$ . Then  $\mathbf{Y}^\top \mathbf{A} \mathbf{Y} = \|\mathbf{Q}_1^\top \mathbf{Y}\|^2$  and  $\mathbf{Q}_1^\top \mathbf{Y} \sim N_r(\mathbf{Q}_1^\top \boldsymbol{\mu}, \mathbf{Q}_1^\top \mathbf{Q}_1) = N_r(\mathbf{Q}_1^\top \boldsymbol{\mu}, \mathbf{I}_r)$ , so by definition  $\mathbf{Y}^\top \mathbf{A} \mathbf{Y} \sim \chi^2(r, \|\mathbf{Q}_1^\top \boldsymbol{\mu}\|^2)$ , and  $\|\mathbf{Q}_1^\top \boldsymbol{\mu}\|^2 = \boldsymbol{\mu}^\top \mathbf{A} \boldsymbol{\mu}$ .

*Necessity.* Apply Theorem 4.2.1 with  $\mathbf{L} = \mathbf{I}$ , so that  $\mathbf{A} = \mathbf{Q} \boldsymbol{\Lambda} \mathbf{Q}^\top$ . Let  $\boldsymbol{\nu} = \mathbf{Q}^\top \boldsymbol{\mu}$ . Then  $\mathbf{b} = \mathbf{Q}^\top \mathbf{A} \boldsymbol{\mu} = \boldsymbol{\Lambda} \boldsymbol{\nu}$ , so  $b_j = \lambda_j \nu_j$ . The constant left after completing squares is  $\boldsymbol{\mu}^\top \mathbf{A} \boldsymbol{\mu} - \sum_{\lambda_j \neq 0} \lambda_j \nu_j^2 = \boldsymbol{\nu}^\top \boldsymbol{\Lambda} \boldsymbol{\nu} - \sum_j \lambda_j \nu_j^2 = 0$ . Hence  $\mathbf{Y}^\top \mathbf{A} \mathbf{Y}$  has the law of  $\sum_{j \in J} \lambda_j (W_j + \nu_j)^2$ , where  $J$  indexes the nonzero eigenvalues.  $J$  is not empty, because a  $\chi^2(r, \gamma)$  variable with  $r \geq 1$  is not identically zero.

By Lemma 4.1.1, near  $t = 0$  the logarithm of the mgf of this sum is  $\sum_{j \in J} \{-\frac{1}{2} \log(1 - 2\lambda_j t) + \lambda_j \nu_j^2 t / (1 - 2\lambda_j t)\}$ . Expanding  $-\frac{1}{2} \log(1 - x) = \sum_{k \geq 1} x^k / (2k)$  and  $x / (1 - x) = \sum_{k \geq 1} x^k$ , the coefficient of  $t^k$  is  $(2^{k-1}/k) \sum_{j \in J} \lambda_j^k (1 + k \nu_j^2)$ . The same expansion of Eq. 4.1.2 gives  $(2^{k-1}/k)(r + k\gamma)$ . Equal distributions have equal mgfs, and two power series that agree on an interval have equal coefficients, so

$$\sum_{j \in J} \lambda_j^k (1 + k \nu_j^2) = r + k\gamma \quad \text{for every } k \geq 1. \quad (4.3.1)$$

Let  $\rho = \max_{j \in J} |\lambda_j|$ . For even  $k$  every term on the left is nonnegative and one of them is at least  $\rho^k$ . The right side grows only linearly in  $k$ , so  $\rho \leq 1$ . Split  $J$  into  $J_+ = \{\lambda_j = 1\}$ ,  $J_- = \{\lambda_j = -1\}$  and  $J_0 = \{0 < |\lambda_j| < 1\}$ , and write  $N_\pm = |J_\pm|$  and  $S_\pm = \sum_{j \in J_\pm} \nu_j^2$ . The terms over  $J_0$  tend to zero as  $k \rightarrow \infty$ , so

$$N_+ + (-1)^k N_- + k(S_+ + (-1)^k S_-) - r - k\gamma \rightarrow 0.$$

Dividing by  $k$  and letting  $k \rightarrow \infty$  through even and through odd values gives  $S_+ + S_- = \gamma = S_+ - S_-$ . Then the same limit without the factor  $k$  gives  $N_+ + N_- = r = N_+ - N_-$ . So  $N_- = S_- = 0$ ,  $N_+ = r$  and  $S_+ = \gamma$ . The terms over  $J_+$  in Eq. 4.3.1 now account for all of  $r + k\gamma$ , and with  $k = 2$  the remaining sum  $\sum_{j \in J_0} \lambda_j^2 (1 + 2\nu_j^2)$  is zero. Every term of that sum is positive, so  $J_0$  is empty. Thus every eigenvalue of  $\mathbf{A}$  is 0 or 1,  $\mathbf{A}$  is idempotent,  $\text{rank}(\mathbf{A}) = N_+ = r$ , and  $\boldsymbol{\mu}^\top \mathbf{A} \boldsymbol{\mu} = \sum_{j \in J_+} \nu_j^2 = \gamma$ .

The necessity half is the one that makes the theorem a tool for *recognizing* distributions. If a quadratic form in a spherical normal vector is chi-squared at all, then it is the squared length of a projection. There is no other way.

### 4.3.2 General covariance matrices

#### Theorem 4.3.2 — When a quadratic form is chi-squared

Let  $\mathbf{Y} \sim N_n(\boldsymbol{\mu}, \boldsymbol{\Sigma})$  and let  $\mathbf{A}$  be symmetric.

- (a) (*Projections.*) If  $\boldsymbol{\Sigma} = \sigma^2 \mathbf{I}$  with  $\sigma > 0$  and  $\mathbf{P}$  is symmetric and idempotent, then

$$\frac{\|\mathbf{P}\mathbf{Y}\|^2}{\sigma^2} = \frac{\mathbf{Y}^\top \mathbf{P} \mathbf{Y}}{\sigma^2} \sim \chi^2(\text{rank}(\mathbf{P}), \frac{\|\mathbf{P}\boldsymbol{\mu}\|^2}{\sigma^2}).$$

- (b) (*Positive definite  $\boldsymbol{\Sigma}$ .*) If  $\boldsymbol{\Sigma}$  is positive definite, then  $\mathbf{Y}^\top \mathbf{A} \mathbf{Y}$  has a  $\chi^2(r, \gamma)$  distribution for some  $r \geq 1$  and  $\gamma \geq 0$  iff  $\mathbf{A}\boldsymbol{\Sigma}$  is idempotent and  $\mathbf{A} \neq \mathbf{0}$ . In that case  $r = \text{rank}(\mathbf{A}) = \text{tr}(\mathbf{A}\boldsymbol{\Sigma})$  and  $\gamma = \boldsymbol{\mu}^\top \mathbf{A} \boldsymbol{\mu}$ .

- (c) (*Any  $\boldsymbol{\Sigma}$ .*) If

$$(i) \ \boldsymbol{\Sigma} \mathbf{A} \boldsymbol{\Sigma} \mathbf{A} \boldsymbol{\Sigma} = \boldsymbol{\Sigma} \mathbf{A} \boldsymbol{\Sigma}, \quad (ii) \ \boldsymbol{\Sigma} \mathbf{A} \boldsymbol{\Sigma} \mathbf{A} \boldsymbol{\mu} = \boldsymbol{\Sigma} \mathbf{A} \boldsymbol{\mu}, \quad (iii) \ \boldsymbol{\mu}^\top \mathbf{A} \boldsymbol{\Sigma} \mathbf{A} \boldsymbol{\mu} = \boldsymbol{\mu}^\top \mathbf{A} \boldsymbol{\mu},$$

then  $\mathbf{Y}^\top \mathbf{A} \mathbf{Y} \sim \chi^2(\text{tr}(\mathbf{A}\boldsymbol{\Sigma}), \boldsymbol{\mu}^\top \mathbf{A} \boldsymbol{\mu})$ . The three conditions hold in particular when  $\mathbf{A}\boldsymbol{\Sigma} \mathbf{A} = \mathbf{A}$ , and when  $\mathbf{A}\boldsymbol{\Sigma}$  is idempotent and  $\boldsymbol{\mu} \in \mathcal{C}(\boldsymbol{\Sigma})$ .

*Proof.*

- (a)  $\sigma^{-1} \mathbf{Y} \sim N_n(\sigma^{-1} \boldsymbol{\mu}, \mathbf{I})$ . Apply Theorem 4.3.1 with  $\mathbf{A} = \mathbf{P}$ , using  $\mathbf{P}^\top \mathbf{P} = \mathbf{P}$  to write  $\mathbf{Y}^\top \mathbf{P} \mathbf{Y} = \|\mathbf{P}\mathbf{Y}\|^2$  and  $\boldsymbol{\mu}^\top \mathbf{P} \boldsymbol{\mu} = \|\mathbf{P}\boldsymbol{\mu}\|^2$ .
- (b) Let  $\boldsymbol{\Sigma}^{1/2}$  be the positive definite square root (Theorem 1.7.6). Then  $\mathbf{Z} = \boldsymbol{\Sigma}^{-1/2} \mathbf{Y} \sim N_n(\boldsymbol{\Sigma}^{-1/2} \boldsymbol{\mu}, \mathbf{I})$  and  $\mathbf{Y}^\top \mathbf{A} \mathbf{Y} = \mathbf{Z}^\top \mathbf{B} \mathbf{Z}$  with  $\mathbf{B} = \boldsymbol{\Sigma}^{1/2} \mathbf{A} \boldsymbol{\Sigma}^{1/2}$  symmetric. By Theorem 4.3.1,  $\mathbf{Y}^\top \mathbf{A} \mathbf{Y}$  is noncentral chi-squared iff  $\mathbf{B}$  is idempotent and nonzero. Now  $\mathbf{B}^2 = \boldsymbol{\Sigma}^{1/2} \mathbf{A} \boldsymbol{\Sigma} \mathbf{A} \boldsymbol{\Sigma}^{1/2}$ , so  $\mathbf{B}^2 = \mathbf{B}$  iff  $\mathbf{A} \boldsymbol{\Sigma} \mathbf{A} = \mathbf{A}$  (multiply by  $\boldsymbol{\Sigma}^{-1/2}$  on both sides) iff  $\mathbf{A} \boldsymbol{\Sigma} \mathbf{A} \boldsymbol{\Sigma} = \mathbf{A} \boldsymbol{\Sigma}$  (multiply by  $\boldsymbol{\Sigma}$  or  $\boldsymbol{\Sigma}^{-1}$  on the right). Also  $\mathbf{B} \neq \mathbf{0}$  iff  $\mathbf{A} \neq \mathbf{0}$ . The degrees of freedom are  $\text{rank}(\mathbf{B}) = \text{rank}(\mathbf{A})$ , which equals  $\text{tr}(\mathbf{B}) = \text{tr}(\mathbf{A}\boldsymbol{\Sigma})$ , and the noncentrality is  $\boldsymbol{\mu}^\top \boldsymbol{\Sigma}^{-1/2} \mathbf{B} \boldsymbol{\Sigma}^{-1/2} \boldsymbol{\mu} = \boldsymbol{\mu}^\top \mathbf{A} \boldsymbol{\mu}$ .
- (c) Take  $\mathbf{L}$  of size  $n \times k$  with full column rank  $k = \text{rank}(\boldsymbol{\Sigma})$  and  $\mathbf{L} \mathbf{L}^\top = \boldsymbol{\Sigma}$ , for instance  $\mathbf{L} = \mathbf{U}_1 \mathbf{D}_1^{1/2}$  from the eigenvectors  $\mathbf{U}_1$  for the positive eigenvalues  $\mathbf{D}_1$  of  $\boldsymbol{\Sigma}$ . (If  $\boldsymbol{\Sigma} = \mathbf{0}$  the claim is trivial.) As in Theorem 4.2.1,  $\mathbf{Y} \stackrel{d}{=} \boldsymbol{\mu} + \mathbf{L} \mathbf{Z}$  with  $\mathbf{Z} \sim N_k(\mathbf{0}, \mathbf{I})$ . Since  $\mathbf{L}$  has full column rank,  $\mathbf{L}^\top$  has full row rank, so  $\mathbf{L} \mathbf{G} \mathbf{L}^\top = \mathbf{L} \mathbf{H} \mathbf{L}^\top$  implies  $\mathbf{G} = \mathbf{H}$ , and  $\mathbf{L} \mathbf{g} = \mathbf{L} \mathbf{h}$  implies  $\mathbf{g} = \mathbf{h}$ . Put  $\mathbf{B} = \mathbf{L}^\top \mathbf{A} \mathbf{L}$  and  $\mathbf{c} = \mathbf{L}^\top \mathbf{A} \boldsymbol{\mu}$ . Condition (i) reads  $\mathbf{L} \mathbf{B}^2 \mathbf{L}^\top = \mathbf{L} \mathbf{B} \mathbf{L}^\top$ , so  $\mathbf{B}^2 = \mathbf{B}$ . Condition (ii) reads  $\mathbf{L} \mathbf{B} \mathbf{c} = \mathbf{L} \mathbf{c}$ , so  $\mathbf{B} \mathbf{c} = \mathbf{c}$ . Condition (iii) reads  $\mathbf{c}^\top \mathbf{c} = \boldsymbol{\mu}^\top \mathbf{A} \boldsymbol{\mu}$ . Hence

$$(\boldsymbol{\mu} + \mathbf{L} \mathbf{Z})^\top \mathbf{A} (\boldsymbol{\mu} + \mathbf{L} \mathbf{Z}) = \boldsymbol{\mu}^\top \mathbf{A} \boldsymbol{\mu} + 2 \mathbf{c}^\top \mathbf{Z} + \mathbf{Z}^\top \mathbf{B} \mathbf{Z} = (\mathbf{Z} + \mathbf{c})^\top \mathbf{B} (\mathbf{Z} + \mathbf{c}),$$

because  $(\mathbf{Z} + \mathbf{c})^\top \mathbf{B} (\mathbf{Z} + \mathbf{c}) = \mathbf{Z}^\top \mathbf{B} \mathbf{Z} + 2 \mathbf{c}^\top \mathbf{B} \mathbf{Z} + \mathbf{c}^\top \mathbf{B} \mathbf{c}$  and  $\mathbf{B} \mathbf{c} = \mathbf{c}$ . Now

$\mathbf{Z} + \mathbf{c} \sim N_k(\mathbf{c}, \mathbf{I})$ , and [Theorem 4.3.1](#) gives the law  $\chi^2(\text{rank } \mathbf{B}, \mathbf{c}^\top \mathbf{B} \mathbf{c})$ , where  $\text{rank } \mathbf{B} = \text{tr } \mathbf{B} = \text{tr}(\mathbf{A}\mathbf{\Sigma})$  and  $\mathbf{c}^\top \mathbf{B} \mathbf{c} = \mathbf{c}^\top \mathbf{c} = \boldsymbol{\mu}^\top \mathbf{A} \boldsymbol{\mu}$ . If  $\mathbf{B} = \mathbf{0}$ , then  $\mathbf{c} = \mathbf{B} \mathbf{c} = \mathbf{0}$  and the form is identically 0, which is  $\chi^2(0)$ . If  $\mathbf{A}\mathbf{\Sigma}\mathbf{A} = \mathbf{A}$ , all three conditions follow by substitution. If  $\mathbf{A}\mathbf{\Sigma}$  is idempotent and  $\boldsymbol{\mu} = \mathbf{\Sigma} \mathbf{v}$ , then (i) follows from multiplying  $\mathbf{A}\mathbf{\Sigma}\mathbf{A}\mathbf{\Sigma} = \mathbf{A}\mathbf{\Sigma}$  by  $\mathbf{\Sigma}$  on the left, and (ii) and (iii) from the same identity applied to  $\mathbf{v}$ .

■

Part (a) is the case used most often in this book. For the linear model  $\mathbf{Y} \sim N_n(\mathbf{X}\boldsymbol{\beta}, \sigma^2 \mathbf{I})$  and the projection  $\mathbf{M}$  onto  $\mathcal{C}(\mathbf{X})$ , [Chapter 6](#) shows that the residual sum of squares is  $\text{SSE} = \|(\mathbf{I} - \mathbf{M})\mathbf{Y}\|^2$ . Since  $(\mathbf{I} - \mathbf{M})\mathbf{X}\boldsymbol{\beta} = \mathbf{0}$ , part (a) gives

$$\text{SSE}/\sigma^2 \sim \chi^2(n - \text{rank}(\mathbf{X})),$$

a central distribution whatever the value of  $\boldsymbol{\beta}$ . The projections that define test statistics, by contrast, do not annihilate the mean under the alternative, and their noncentrality  $\|\mathbf{P}\mathbf{X}\boldsymbol{\beta}\|^2/\sigma^2$  measures how far the truth is from the hypothesis in the metric that the test can see.

In part (c) idempotence of  $\mathbf{A}\mathbf{\Sigma}$  alone is *not* enough when  $\mathbf{\Sigma}$  is singular. A mean with a component outside  $\mathcal{C}(\mathbf{\Sigma})$  is a deterministic shift that the form may pick up as a constant or as a linear term ([Exercise \(B2\)](#)). Part (c) also covers the natural generalization of the Mahalanobis distance.

### Corollary 4.3.3 — Mahalanobis forms

Let  $\mathbf{Y} \sim N_n(\boldsymbol{\mu}, \mathbf{\Sigma})$ .

- (a) If  $\mathbf{\Sigma}$  is positive definite,  $(\mathbf{Y} - \boldsymbol{\mu})^\top \mathbf{\Sigma}^{-1}(\mathbf{Y} - \boldsymbol{\mu}) \sim \chi^2(n)$  and  $\mathbf{Y}^\top \mathbf{\Sigma}^{-1} \mathbf{Y} \sim \chi^2(n, \boldsymbol{\mu}^\top \mathbf{\Sigma}^{-1} \boldsymbol{\mu})$ .
- (b) If  $\mathbf{G}$  is a symmetric generalized inverse of  $\mathbf{\Sigma}$  and  $\boldsymbol{\mu} \in \mathcal{C}(\mathbf{\Sigma})$ , then  $\mathbf{Y}^\top \mathbf{G} \mathbf{Y} \sim \chi^2(\text{rank}(\mathbf{\Sigma}), \boldsymbol{\mu}^\top \mathbf{G} \boldsymbol{\mu})$ .

*Proof.*

- (a)  $\mathbf{A} = \mathbf{\Sigma}^{-1}$  satisfies  $\mathbf{A}\mathbf{\Sigma}\mathbf{A} = \mathbf{A}$ , and  $\mathbf{Y} - \boldsymbol{\mu} \sim N_n(\mathbf{0}, \mathbf{\Sigma})$ . The central statement was already obtained by whitening in [Proposition 3.2.3](#).
- (b)  $\mathbf{G}\mathbf{\Sigma}$  is idempotent because  $\mathbf{\Sigma}\mathbf{G}\mathbf{\Sigma} = \mathbf{\Sigma}$ , so part (c) of the theorem applies and the degrees of freedom are  $\text{tr}(\mathbf{G}\mathbf{\Sigma}) = \text{tr}(\mathbf{L}^\top \mathbf{G} \mathbf{L})$ . With  $\mathbf{L}$  as in that proof,  $\mathbf{L}\mathbf{L}^\top \mathbf{G} \mathbf{L}\mathbf{L}^\top = \mathbf{L}\mathbf{L}^\top$  gives  $\mathbf{L}^\top \mathbf{G} \mathbf{L} = \mathbf{I}_k$ , whose trace is  $k = \text{rank}(\mathbf{\Sigma})$ .

■

A symmetric generalized inverse always exists ([Proposition 1.3.3\(d\)](#)). The value of  $\mathbf{Y}^\top \mathbf{G} \mathbf{Y}$  does not depend on which one is chosen, with probability one ([Exercise \(B3\)](#)).

#### Example. (Correlated observations about a common level)

Let  $\mathbf{Y} \sim N_n(\boldsymbol{\mu}, \mathbf{\Sigma})$  with  $\mathbf{\Sigma}$  positive definite, and suppose we want to measure how far  $\mathbf{Y}$  is from a common level  $m\mathbf{1}$ . The generalized least squares level is  $\hat{m} = \mathbf{1}^\top \mathbf{\Sigma}^{-1} \mathbf{Y} / \mathbf{1}^\top \mathbf{\Sigma}^{-1} \mathbf{1}$ ,

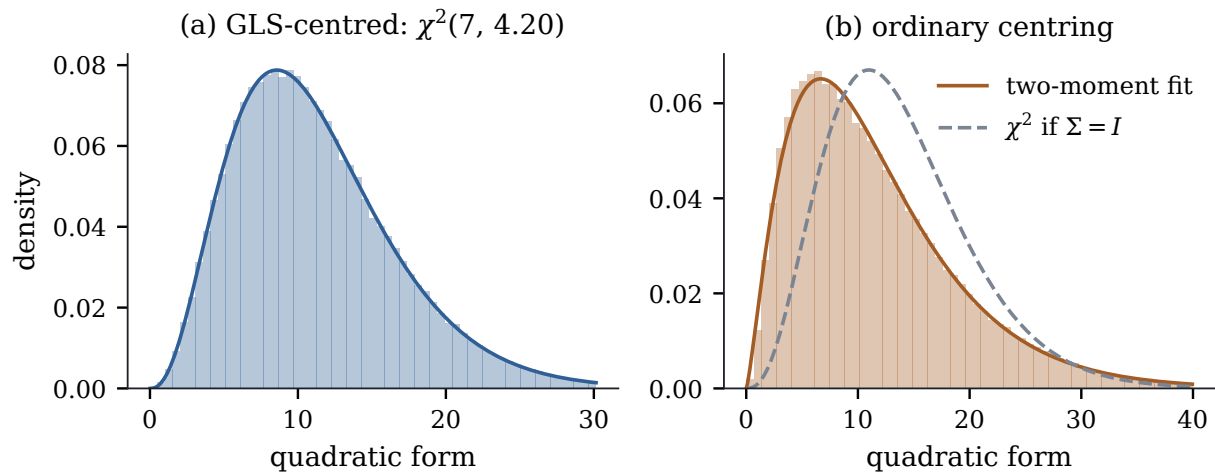


Figure 4.3.1: Two quadratic forms measuring spread about a common level, for  $n = 8$  autoregressive observations with a trending mean. (a) With generalized least squares centring,  $\mathbf{A}\Sigma$  is idempotent and the form is exactly noncentral chi-squared. (b) With ordinary centring it is a weighted sum of noncentral  $\chi^2(1)$  terms. A two-moment scaled chi-squared fits closely. The chi-squared law that would hold for independent observations does not.

and the natural discrepancy is  $(\mathbf{Y} - \hat{m}\mathbf{1})^\top \Sigma^{-1}(\mathbf{Y} - \hat{m}\mathbf{1}) = \mathbf{Y}^\top \mathbf{A} \mathbf{Y}$  with

$$\mathbf{A} = \Sigma^{-1} - \frac{\Sigma^{-1} \mathbf{1} \mathbf{1}^\top \Sigma^{-1}}{\mathbf{1}^\top \Sigma^{-1} \mathbf{1}}.$$

A direct multiplication gives  $\mathbf{A}\Sigma\mathbf{A} = \mathbf{A}$ , and  $\text{tr}(\mathbf{A}\Sigma) = n - 1$ . By Theorem 4.3.2(b) the form is noncentral chi-squared with  $n - 1$  degrees of freedom and noncentrality  $\boldsymbol{\mu}^\top \mathbf{A} \boldsymbol{\mu}$ , which is zero exactly when  $\boldsymbol{\mu}$  is constant. The listing takes  $n = 8$ , an autoregressive correlation  $\Sigma = (\rho^{|i-j|})$  with  $\rho = 0.6$ , and a linear trend in the mean, for which  $\gamma = 4.200$ . Figure 4.3.1(a) compares 200,000 simulated values with the  $\chi^2(7, 4.200)$  density.

```
import numpy as np
from scipy import integrate, stats
rng = np.random.default_rng(31415)
n, rho = 8, 0.6
idx = np.arange(n)
Sigma = rho ** np.abs(idx[:, None] - idx[None, :]) # AR(1) correlation matrix
mu = 0.4 * (idx - idx.mean()) # linear trend, mean zero
one = np.ones(n)

W = np.linalg.inv(Sigma)
A = W - np.outer(W @ one, W @ one) / (one @ W @ one) # GLS-centred form
AS = A @ Sigma
print("A Sigma idempotent:", np.allclose(AS @ AS, AS))
df, gamma = np.trace(AS), mu @ A @ mu # chi^2(df, gamma)

L = np.linalg.cholesky(Sigma) # Y = mu + L Z
Z = rng.standard_normal((200_000, n))
Y = mu + Z @ L.T
q_gls = np.einsum("ij,jk,ik->i", Y, A, Y)
print(df, gamma, stats.kstest(q_gls, stats.ncx2(df, gamma).cdf).pvalue)
```

[Run this code online](#)

### 4.3.3 Forms that are not chi-squared

Most quadratic forms fail the conditions of [Theorem 4.3.2](#). The ordinary sum of squares about the mean,  $\mathbf{Y}^\top \mathbf{C} \mathbf{Y}$  with  $\mathbf{C} = \mathbf{I} - n^{-1} \mathbf{1} \mathbf{1}^\top$ , is an example as soon as the observations are correlated. Then  $\mathbf{C} \boldsymbol{\Sigma}$  is not idempotent, and by Theorem 4.2.1 the law is that of  $\sum_j \lambda_j (W_j + \nu_j)^2$ , with weights  $\lambda_j$  equal to the nonzero eigenvalues of  $\mathbf{C} \boldsymbol{\Sigma}$ . In [Example \(Correlated observations about a common level\)](#) these range from 0.259 to 1.832. Two practical ways to handle such a law are the following.

**Exact computation.** The characteristic function of the weighted sum is known in closed form (put  $t = iu$  in Corollary 4.2.2), and a distribution function can be recovered from a characteristic function  $\varphi$  by the inversion formula of Gil-Pelaez (1951),

$$\Pr(X > x) = \frac{1}{2} + \frac{1}{\pi} \int_0^\infty \frac{\operatorname{Im}\{e^{-iux} \varphi(u)\}}{u} du.$$

For a weighted sum of  $k$  chi-squared terms the integrand decays like  $u^{-1-k/2}$ , so ordinary quadrature works well. Imhof (1961) gave the real-variable form of this integral that most software uses. The listing implements it directly and checks it against simulation.

**Two-moment approximation.** Match  $a\chi^2(\nu)$  to the mean  $m$  and variance  $v$  from Theorem 4.2.3. Since  $a\chi^2(\nu)$  has mean  $a\nu$  and variance  $2a^2\nu$ , this gives  $a = v/(2m)$  and  $\nu = 2m^2/v$ , with  $\nu$  usually not an integer (Satterthwaite 1946; Patnaik 1949).

In [Example \(Correlated observations about a common level\)](#), the simulated 95th percentile of  $\mathbf{Y}^\top \mathbf{C} \mathbf{Y}$  is 26.51. The exact tail probability beyond it is 0.0490, and the approximation with  $a = 2.488$  and  $\nu = 4.68$  gives 0.0484. Treating the observations as independent, and so using  $\chi^2(7, 6.720)$ , gives 0.040. [Figure 4.3.1\(b\)](#) shows the three laws. Here the error from ignoring correlation is modest at the 5% point but visible in the shape. It is not a monotone function of  $\rho$ , but strong correlation makes it severe: with  $\rho = 0.9$  and the same mean, the naive tail probability at the true 95th percentile is 0.197.

```
def canonical(A, mu, L):
    """Y^T A Y = sum_j lam_j (W_j + nu_j)^2 + c (all lam_j nonzero here), W ~ N(0, I)
    """
    lam, Qm = np.linalg.eigh(L.T @ A @ L)
    b = Qm.T @ L.T @ A @ mu
    keep = np.abs(lam) > 1e-10
    assert np.allclose(b[~keep], 0) # no linear terms in this
    example
    c = mu @ A @ mu - np.sum(b[keep] ** 2 / lam[keep])
    return lam[keep], b[keep] / lam[keep], c

def tail_prob(x, lam, nu, c=0.0):
    """P(sum lam_j (W_j + nu_j)^2 + c > x) by Gil-Pelaez inversion."""
    def integrand(u):
        z = 1 - 2j * lam * u
        phi = np.prod(z ** -0.5 * np.exp(1j * u * lam * nu**2 / z))
        return (np.exp(-1j * u * (x - c)) * phi).imag / u
    edges = np.linspace(0, 400, 401) # integrand decays like u
    ^(-1-k/2)
    val = sum(integrate.quad(integrand, a, b)[0] for a, b in zip(edges[:-1], edges
    [1:]))
    return 0.5 + val / np.pi
```

[Run this code online](#)

## 4.3.4 Exercises

### 4.3.4.1 A. Check your understanding

#### Exercise A1.

Let  $\mathbf{Y} \sim N_n(\mathbf{0}, \mathbf{I})$  with  $n \geq 2$ , and  $\mathbf{A} = \alpha \mathbf{I} + \beta \mathbf{1}\mathbf{1}^\top$ . Find all  $(\alpha, \beta)$  for which  $\mathbf{Y}^\top \mathbf{A} \mathbf{Y}$  has a chi-squared distribution, and identify each form.

*Solution.*  $\mathbf{A}$  has eigenvalue  $\alpha + n\beta$  on  $\mathbf{1}$  and  $\alpha$  on  $\mathbf{1}^\perp$ . By [Theorem 4.3.1](#), the form is chi-squared iff  $\mathbf{A} \neq \mathbf{0}$  and both eigenvalues lie in  $\{0, 1\}$ . The three solutions are  $(1, 0)$ , giving  $\|\mathbf{Y}\|^2 \sim \chi^2(n)$ ;  $(0, 1/n)$ , giving  $n\bar{Y}^2 \sim \chi^2(1)$ ; and  $(1, -1/n)$ , giving  $\sum (Y_i - \bar{Y})^2 \sim \chi^2(n-1)$ .

#### Exercise A2.

Two independent samples of sizes  $m$  and  $k$  come from  $N(\mu_1, \sigma^2)$  and  $N(\mu_2, \sigma^2)$ . Write the pooled within-sample sum of squares as  $\|(\mathbf{I} - \mathbf{M})\mathbf{Y}\|^2$  for a suitable projection, and show that the two-sample  $t$  statistic has the  $t(m+k-2, (\mu_1 - \mu_2)/(\sigma\sqrt{1/m+1/k}))$  distribution.

### 4.3.4.2 B. Practice

#### Exercise B1.

Let  $\mathbf{Y} \sim N_n(\mu\mathbf{1}, \sigma^2\{(1-\rho)\mathbf{I} + \rho\mathbf{1}\mathbf{1}^\top\})$  with  $-1/(n-1) < \rho < 1$ , and let  $\bar{Y}$  and  $S^2$  be the sample mean and variance. Show that  $(n-1)S^2/\{\sigma^2(1-\rho)\} \sim \chi^2(n-1)$  and that  $S^2$  is independent of  $\bar{Y}$ . Deduce that  $\sqrt{n}(\bar{Y} - \mu)/S$  has the law of  $\{(1 + (n-1)\rho)/(1-\rho)\}^{1/2}$  times a  $t(n-1)$  variable, and describe what a positive  $\rho$  does to the coverage of the usual 95%  $t$  interval for  $\mu$ .

#### Exercise B2.

Let  $\Sigma = \text{diag}(1, 0)$ ,  $\mu = (0, 1)^\top$ ,  $\mathbf{A} = \mathbf{I}_2$  and  $\mathbf{Y} \sim N_2(\mu, \Sigma)$ . Show that  $\mathbf{A}\Sigma$  is idempotent but  $\mathbf{Y}^\top \mathbf{A} \mathbf{Y}$  is not chi-squared. Which of the conditions of [Theorem 4.3.2\(c\)](#) fail? What changes if  $\mu = (m, 0)^\top$ ? Repeat the first part with  $\mathbf{A} = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}$ , where the shift enters through a linear term.

*Solution.*  $\mathbf{A}\Sigma = \Sigma$  is idempotent. But  $Y_2 = 1$  with probability one, so  $\mathbf{Y}^\top \mathbf{A} \mathbf{Y} = Y_1^2 + 1 \geq 1$ , while every  $\chi^2(r, \gamma)$  with  $r \geq 1$  gives positive probability to  $(0, 1)$ . Conditions (i) and (ii) hold, since  $\Sigma^3 = \Sigma$  and  $\Sigma^2\mu = \mathbf{0} = \Sigma\mu$ . Condition (iii) fails, since  $\mu^\top \Sigma\mu = 0 \neq 1 = \mu^\top \mu$ . The part of the mean outside  $\mathcal{C}(\Sigma)$  adds a constant. If  $\mu = (m, 0)^\top \in \mathcal{C}(\Sigma)$ , then  $\mathbf{Y}^\top \mathbf{A} \mathbf{Y} = Y_1^2 \sim \chi^2(1, m^2)$ , as [Theorem 4.3.2\(c\)](#) predicts. For  $\mathbf{A} = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}$  and  $\mu = (0, 1)^\top$ ,  $\mathbf{A}\Sigma = \begin{pmatrix} 1 & 0 \\ 1 & 0 \end{pmatrix}$  is again idempotent, and  $\mathbf{Y}^\top \mathbf{A} \mathbf{Y} = Y_1^2 + 2Y_1Y_2 = Y_1^2 + 2Y_1 = (Y_1 + 1)^2 - 1$ , which is negative with positive probability. Conditions (i) and (ii) hold, since  $\Sigma\mathbf{A}\Sigma = \Sigma$  and  $\Sigma\mathbf{A}\Sigma\mu = \Sigma\mu = (1, 0)^\top$ , but (iii) fails because  $\mu^\top \mathbf{A}\Sigma\mu = 1 \neq 0 = \mu^\top \mu$ . The shift now enters through the linear term  $2Y_1$ .

**Exercise B3.**

In [Corollary 4.3.3\(b\)](#), show that  $\mathbf{Y}^\top \mathbf{G} \mathbf{Y}$  takes the same value for every generalized inverse  $\mathbf{G}$  of  $\mathbf{\Sigma}$ , with probability one. *Hint:* Theorem 2.2.3.

*Solution.* By Theorem 2.2.3,  $\mathbf{Y} - \boldsymbol{\mu} \in \mathcal{C}(\mathbf{\Sigma})$  with probability one. Since  $\boldsymbol{\mu} \in \mathcal{C}(\mathbf{\Sigma})$ , also  $\mathbf{Y} \in \mathcal{C}(\mathbf{\Sigma})$ , so  $\mathbf{Y} = \mathbf{\Sigma} \mathbf{v}$  for some (random)  $\mathbf{v}$ . For any generalized inverse  $\mathbf{G}$ ,  $\mathbf{Y}^\top \mathbf{G} \mathbf{Y} = \mathbf{v}^\top \mathbf{\Sigma} \mathbf{G} \mathbf{\Sigma} \mathbf{v} = \mathbf{v}^\top \mathbf{\Sigma} \mathbf{v}$ , which does not involve  $\mathbf{G}$ .

**Exercise B4.**

Let  $\mathbf{Y} \sim N_n(\mathbf{0}, \mathbf{\Sigma})$  with  $\mathbf{\Sigma}$  positive definite, and let  $\mathbf{A} \neq \mathbf{0}$  be nonnegative definite. Show that the two-moment approximation  $a\chi^2(\nu)$  to  $\mathbf{Y}^\top \mathbf{A} \mathbf{Y}$  has  $\nu = \{\text{tr}(\mathbf{A} \mathbf{\Sigma})\}^2 / \text{tr}\{(\mathbf{A} \mathbf{\Sigma})^2\}$ , and that  $\nu \leq \text{rank}(\mathbf{A})$ , with equality iff all nonzero eigenvalues of  $\mathbf{A} \mathbf{\Sigma}$  are equal.

**4.3.4.3 C. Going deeper****Exercise C1.**

Let  $\mathbf{Y} \sim N_n(\boldsymbol{\mu}, \mathbf{\Sigma})$  with  $\mathbf{\Sigma}$  possibly singular, and suppose  $\mathbf{Y}^\top \mathbf{A} \mathbf{Y} \sim \chi^2(r, \gamma)$  with  $r \geq 1$ . Show that  $\mathbf{\Sigma} \mathbf{A} \mathbf{\Sigma} \mathbf{A} \mathbf{\Sigma} = \mathbf{\Sigma} \mathbf{A} \mathbf{\Sigma}$ . *Hint:* in [Corollary 4.2.2](#), compare the coefficients of  $t^k$  for  $k \geq 3$  with those of [Eq. 4.1.2](#), and adapt the argument of [Theorem 4.3.1](#).